

- **android.permission.NFC** declares that the particular app would require access to the Cardboard's NFC tag.
- **android.permission.READ_EXTERNAL_STORAGE** declares that the app would like to pair the user's phone to the VR viewer.
- The intent-filter and specifically **com.google.intent.category.CARDBOARD** declares that the current activity is compatible with VR viewers.
- **<uses-feature android:glEsVersion="0x00020000" android:required="true" />** declares that the device must be able to support OpenGL ES 2.0 to run this app.
- **android:screenOrientation="landscape"** declares that the activity's screen orientation must be "landscape." It is mandatory for VR Apps to render on fullscreen and landscape and thus have landscape orientation.
- **android:configChanges="orientation|keyboardHidden"** declaration is not mandatory, but recommended.

8.5 Extend CardboardActivity

CardboardActivity defines the base for coding a cardboard app. It is the base activity that lets an app integrate with Cardboard devices. It exposes events to interact with Cardboards and handles many of the details commonly required when creating an activity for VR rendering. The **CardboardActivity** is based on sticky immersive mode i.e. it makes the content take up the whole screen, hiding the system UI.

Define a CardboardView

All the UI elements in the Android app are built using views. The Android Cardboard SDK provides the developers with its own view, **CardboardView**, that can be used for VR rendering. Check the example below to find out how **CardboardView** is defined in the activity layout xml file :

```
<com.google.vrtoolkit.cardboard.CardboardView
    android:id="@+id/cardboard_view
    android:layout_width="fill_parent"
    android:layout_height="fill_parent" />
```

Now, in the main activity class initialize the **CardboardView** in the **onCreate()** method:

Conclusion

The market share of Android users in the smartphone segment is unpar-